Problem Set on Transformers

1. $I_1 = \dfrac{250\text{ kVA}}{11\text{ kV}} = \mathbf{22.7A}; I_2 = \dfrac{250\text{ kVA}}{415\text{ V}} = \mathbf{602\,A};\ N_1 = \dfrac{N_2}{K} = 2121;\ \Phi_m = \dfrac{415}{4.44\times 50\times 80} = \mathbf{23.4\,mWb}$

2. $\Phi_m = \dfrac{6400}{4.44\times 50\times 480} = \mathbf{0.06\ Wb};\ E_2 = 4.44\times 50\times 20\times 0.06 = \mathbf{266.4\ V}$

3. $(a)\, K = 400/80 = 5; E_2 = 5\times 240 = \mathbf{1200\,V};\ (b)\, B_m = \dfrac{\Phi_m}{A} = \dfrac{240}{4.44\times 50\times 80\times 200\times 10^{-4}} = \mathbf{0.675\,T}$

4. $(a)\, N_1 = \dfrac{V_1}{V_2}\times N_2 = \dfrac{2000}{240}\times 60 = \mathbf{500}; (b) I_1 = \dfrac{10\,\text{kVA}}{2000\,\text{V}} = \mathbf{5\,A}, I_2 = \dfrac{10\,\text{kVA}}{240\,\text{V}} = \mathbf{41.67\,A}$

5. $(a)\, I_1 = \dfrac{200\,000}{3300} = \mathbf{60.6\,A}; I_2 = \dfrac{200\,000}{240} = \mathbf{833.33\,A}; (b)\, \Phi_m = \dfrac{240}{4.44\times 50\times 80} = \mathbf{0.0135\ Wb}; (c)\, N_1 = 80\times\dfrac{3300}{240} = \mathbf{1100}$

6. $I_2 = \dfrac{50\,000}{240} = 208.33\,A;\ Z = \dfrac{240}{208.33} = \mathbf{1.152\,\Omega};\ \Phi_m = \dfrac{240}{4.44\times 50\times 23} = \mathbf{0.047\,Wb}$

7. $\phi_0 = \cos^{-1} 0.4 = 66.42°,\ \phi_2 = \cos^{-1} 0.8 = 36.87°$**;** $I_1^{'} = (1/2)\times 50 = 25\text{ A};\ \theta = 66.42° - 36.87° = 29.55°$

$I_1 = \sqrt{I_0^2 + I_1^{'2} + 2I_0 I_1^{'}\cos 29.55°} = \mathbf{25.9\ A}$

8. $(a)\, E_2 = \left(\dfrac{400}{100}\right)\times 230 = \mathbf{920\,V}; (b)\, \Phi_m = \dfrac{230}{4.44\times 50\times 100} = 0.01036,\ B_m = \dfrac{0.01036}{250\times 10^{-4}} = \mathbf{0.414\ T}$

9. $(a)\, \Phi_m = \dfrac{230}{4.44\times 50\times 30};\ B_m = \dfrac{0.03453}{0.025} = \mathbf{1.3814\ T}; (b)\, E_2 = K\times 230 = \mathbf{2683.33\,V}; (c)\, I_1 = KI_2 = \mathbf{1166.67\ A}$

10. $\cos\phi_0 = \dfrac{P_0}{V_1 I_0} = 0.4557;\ \phi_0 = \cos^{-1} 0.4557 = 62.9°; I_m = I_0 \sin\phi_0 = \mathbf{0.5697\,A}; I_w = I_0\cos\phi_0 = \mathbf{0.2917\,A}$

$\mathbf{I}_1^{'} = K\mathbf{I}_2 = 4\times 0.8 - j4\times 0.6 = (3.2 - j2.4)\text{ A};\ \mathbf{I}_0 = (0.2917 - j0.5697)\text{ A};\ \mathbf{I}_1 = \mathbf{I}_1^{'} + \mathbf{I}_0 = \mathbf{4.584}\angle 40.38°\,\mathbf{A};$

$pf = \cos 40.38° = \mathbf{0.762}$

11. $\Phi_m = \dfrac{220}{4.44\times 50\times 100};\ A = \dfrac{\Phi_m}{B_m} = \mathbf{8.26\times 10^{-3}\,m^2}$**;** $(a)\, V_2 = \left(\dfrac{N_2}{N_1}\right)V_1 = \mathbf{440\ V}$**;** $(b)\, \phi = \cos^{-1} 0.8,\ \mathbf{I}_2 = I_2(\cos\phi - j\sin\phi)$

$\mathbf{I}_2 = (6.4 - j4.8),\ \mathbf{I}_1 = -\left(\dfrac{N_2}{N_1}\right)\mathbf{I}_2 = -2(6.4 - j4.8) = \mathbf{16\angle 143.1°\ A}$

12. $(a)\ N_1 = \left(\dfrac{11000}{400}\right)\times 100 = \mathbf{2750};\ (b)\, I_1 = \dfrac{500\,\text{kVA}}{11\text{kV}} = \mathbf{45.45\ A},\ I_2 = \dfrac{500\,\text{kVA}}{400\text{ V}} = \mathbf{1250\ A};$

$(c)\ \Phi_m = \dfrac{400}{4.44\times 50\times 100} = \mathbf{0.018\ Wb}$